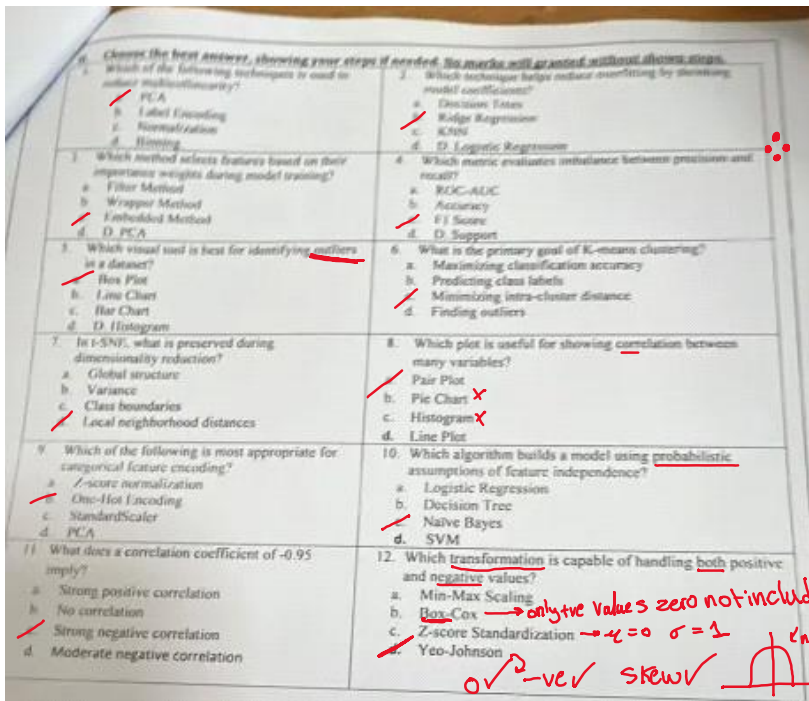




Question # 2 (ILO A3, A8 OUTCOME 6): (8 marks)

An AI tool diagnoses patients into four possible conditions: "Healthy", "Flu", "COVID-19", and "Pneumonia". The results for 240 patients are summarized below:
 (4 marks)

	Pred. Healthy	Pred. Flu	Pred. COVID-19	Pred. Pneumonia
Actual Healthy	60	10	1	0
Actual Flu	5	40	1	2
Actual COVID-19	2	3	35	10
Actual Pneumonia	0	2	5	17

Handwritten notes: **Actual** (vertical), **Predicted** (horizontal), **P=TP** (circled), **Q=FP** (circled).

a. Calculate the overall accuracy of the model.

b. Compute the precision and recall for the "COVID-19" class.

c. Which class has the lowest recall? What are the real-world risks of low recall for that class?

d. State whether the dataset is balanced or not and mention how it may affect the results

$$a) = \frac{TN+TP}{TN+TP+FP+FN} \rightarrow \frac{60+48+35+17}{\text{everything}} = \checkmark$$

$$b) \text{ Precision} \rightarrow \frac{TP}{TP+FP} = \frac{35}{35+1+6+5} = \checkmark$$

$$\text{sensitivity (Recall (TPR))} \rightarrow \frac{TP}{TP+FN} = \frac{35}{35+2+3+10} = \checkmark$$

c) Pneumonia. People with it can be treated for something else due to high false negatives

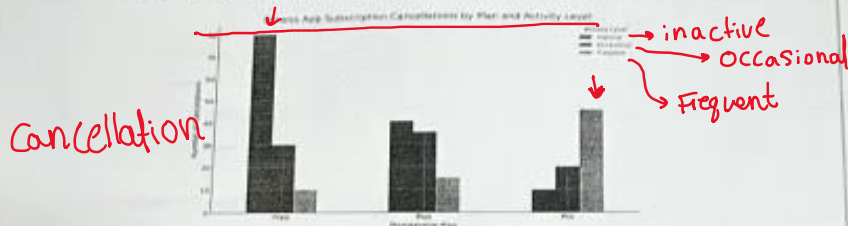
How it affects the results:

- Classifier Bias:** Machine learning models trained on imbalanced data naturally tend to optimize for the majority classes (Healthy and Flu) to achieve higher overall accuracy. It learns the features of the majority classes much better than the minority ones.
- Misleading Accuracy:** The high accuracy score (66.67%) is heavily carried by the high performance on the "Healthy" class (92.31% recall). It masks the fact that the model performs poorly on the critical, rarer condition like Pneumonia (68% recall).

Attempt all the following questions
Question #1 (ILO AS OUTCOME 6) (8 marks)

A fitness app monitors cancellations by marks)

- Membership Plan: Free, Plus, Pro
- User Activity Level: Inactive, Occasional, Frequent

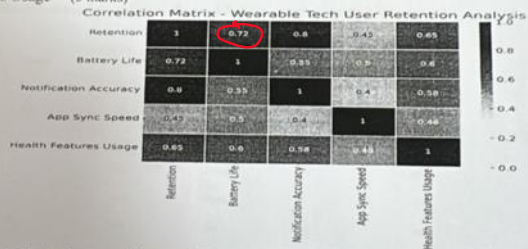


- Identify any unexpected patterns. What might be some causes behind this observation?
- Which segment exhibits the highest churn, and how could the app respond strategically?

a) Pro users who are interacting frequently still cancel alot, this might be due to app not meeting expectations, overpricing, inaccuracy.

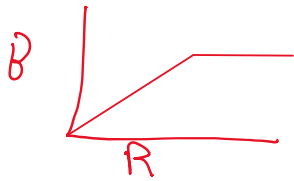
b) Free → inactive. The app can refresh its UI or send notifications about its change. Survey people about what can be done better

II. A wearable tech firm analyzes data to assess what features improve monthly user retention. Metrics include Retention, Battery Life, Notification Accuracy, App Sync Speed, Health Features Usage (5 marks)



- Compare the influence of Battery Life and Health Features Usage on Retention.
- There's a 0.80 correlation between Notification Accuracy and Retention. Does this suggest a direct cause-effect? Explain.
No, correlation is not causation. A 3rd factor can be influential too.
- A small number of users with poor sync speeds and 0 retention may skew the results. What preprocessing steps should be taken before modeling?
z-score, IQR, Segmentation - filtering, robust
- A plot of Battery Life vs Retention shows a linear trend that flattens beyond 5 days. What type of relationship might this indicate?

a) Battery Life - Retention → 0.72
Health Features - Retention → 0.65
this means that the higher the battery life the higher the retention due to it having a higher positive correlation.



non-linear relation

Meaning: Up to 5 days, every extra day of battery life significantly increases a user's likelihood to stay retained. However, past the 5-day mark, the user is already fully satisfied with the battery performance. Adding even more battery life (e.g., making it last 7 or 10 days) provides no extra practical value to the user experience and stops moving the needle on retention.

Question 2.3: BLOA OUTCOME 6: (8 marks)

Find the least squares regression line for the data on incomes and food expenditures of the seven households given in the table below. Use income as an independent variable and food expenditure as a dependent variable such that $\hat{y} = a + bx$.

Household	Income (x)	Food Expenditure (y)
1	12	1.4
2	15	2.4
3	18	1.5
4	20	2.0
5	22	1.9
6	25	1.5
7	28	1.7

a) Compute $\sum x$, $\sum y$, $\sum xy$, $\sum x^2$

$\sum x$	388
$\sum y$	10.8
$\sum xy$	55.428
$\sum x^2$	23058

b) Compute SS_y and SS_x

SS_y	$6403 - \frac{388 \cdot 10.8}{7} = 416.714$	SS_x	$23058 - \frac{388^2}{7} = 1551.714$
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$$SS_{xx} = \sum (x^2) - \frac{[\sum (x)]^2}{n}$$

$$SS_{xy} = [\sum x \cdot y] - \frac{[\sum x] \cdot [\sum y]}{n}$$

$$TSS = SS_{yy} \rightarrow \sum (y^2) - \frac{[\sum (y)]^2}{n}$$

$$SSE = \sum (y - \hat{y})^2$$

$$SSR = TSS - SSE$$

$$\hat{y} = b_0 + b_1 x$$

$\hat{y} - b_1 x$ $\frac{SS_{xy}}{SS_{xx}}$

$$b_1 = \frac{416.714}{1551.714} = 0.2685$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$b_0 = 15.428 - 0.2685(55.428)$$

$$b_0 = 0.5455$$

$$\hat{y} = 0.5455 + 0.2685x$$

Analyze the following dataset below shows data for students preparing for an exam. (4 marks)

Student	Study Time	Sleep	Attended Lectures	Passed
1	Long	Enough	Yes	Yes
2	Short	Low	No	No
3	Long	Low	Yes	Yes
4	Short	Enough	Yes	No
5	Long	Enough	No	Yes
6	Short	Low	Yes	No
7	Long	Low	No	No
8	Long	Enough	Yes	Yes
9	Short	Enough	No	No
10	Long	Low	Yes	Yes

Study Time = Long, Sleep = Low, attended = Yes using Naive Bayes. Show all intermediate

Studytime	Yes	No
Long	5/5	1/5
Short	0	4/5

Sleep	Yes	No
Enough	3/5	2/5
Low	2/5	3/5

Lec	Yes	No
yes	4/5	2/5
No	1/5	3/5

$$\frac{5}{10} \rightarrow \text{yes} \quad \frac{5}{10} \rightarrow \text{No}$$

$$\text{Yes} \rightarrow 5 \quad \text{No} \rightarrow 5$$

$$P(\text{Yes}) \rightarrow \frac{5}{5} \cdot \frac{2}{5} \cdot \frac{4}{5} \cdot \frac{5}{10} = 0.16$$

$$P(\text{No}) \rightarrow \frac{1}{5} \cdot \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{5}{10} = 0.024$$

Yes > No $\rightarrow$ therefore Passed ✓

yes > No $\rightarrow$ the sort passed